



AI Strategy for I&E LLC

KI-Strategie

From a manual document check to a human-supervised AI co-pilot that checks every shipment before it leaves.

Author	Janaina Hoffmann , AI Consultant, Berlin
Company (fictional)	I&E LLC, importer/exporter of construction materials, Germany · CEO: Chleo
Version	1.0 · 4 October 2026 · language: English (confirmed for the exam)
Live system	Streamlit MVP + n8n workflows + LangSmith evaluation (repository: github.com/Ninagfcosta/ironhack-capstone_klarschiff)
Related documents	use_case_definition.md · roi_risk_assessment.md · strategic_plan.md · compliance/ · presentation.pdf · one-page summary

Scenario and figures are fictional or illustrative (training project). Decision support only: a qualified person confirms every classification and document before any customs declaration. This document follows the IHK requirement: "Als Projektarbeit ist eine KI-Strategie zu erstellen", submitted digitally before the oral exam, presented in 10–15 minutes and discussed in a 10–15 minute Fachgespräch.

Contents and link to the IHK modules

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1 Executive summary

Recommendation: start a **3-month paid pilot** (one route, one product family, a blind test first), then decide go / no-go. Full production after about **12 months** in three phases with a gate after each phase.

- **Problem.** I&E checks invoices, packing lists and certificates by hand. One missing document (for example the EU Declaration of Performance) stops goods at the border: storage fees, late customers, stress. Rules change fast (CBAM since January 2026, the EU steel measure since July 2026, EUDR from December 2026).
- **AI goal.** Every shipment is checked **before it leaves**. The AI does the searching and checking; people keep the decisions.
- **Use case.** Chosen from five candidates with a value/feasibility matrix: the **pre-shipment document check with customs-code (HS) suggestion**. High value, feasible with today's data, low legal risk.
- **Solution.** A five-step agent: Intake → Validate → Recommend (RAG + GPT-4o-mini) → Rules (versioned, with legal sources) → a person decides. Built and tested in the bootcamp (n8n POC, Python agent, Streamlit app, LangSmith).
- **Value.** Base case: about €6,400 per month, break-even in month 7, +60% ROI after 36 months (illustrative, measured in the pilot).
- **Risk.** EU AI Act: **minimal risk** (with the Art. 50 AI disclosure for the assistant). GDPR: only goods data goes to the AI. Main risk is a wrong code accepted without checking; controls: confidence threshold, 13 review triggers, human approval, decision log.

2 Starting point and AI readiness

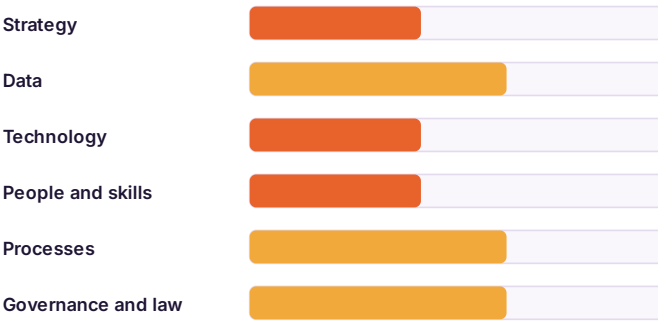
M3 · M5

Company profile

- SME, construction-materials import/export, Germany; imports from Türkiye, Serbia, China; occasional exports to the US.
- 2–3 logistics staff; an external customs broker files the declaration in ATLAS and carries legal responsibility.
- No in-house IT team; standard office and cloud tools.

Current process (as-is)

- One person checks documents by hand, no checklist, no second review.
- 30–45 minutes per shipment; errors are found only at the border.
- Rules differ by product (CPR, CBAM, steel measure, REACH, EUDR) and change often.



AI readiness assessment (1 = low, 5 = high). Own assessment, fictional company.

Dimension	Score	Evidence	Consequence for the strategy
Strategy	2	No AI strategy, but a CEO sponsor	This document; pilot with clear go / no-go
Data	3	Digital invoices; e-invoices coming (DE: receive 2025, issue 2027/28)	Read e-invoices first; build a product master list in the pilot
Technology	2	No IT team, SaaS only	Light cloud app, no deep integration before Phase 2
People	2	No AI experience	AI-literacy training, 2 key users in the design
Processes	3	Repeatable manual check	Standardise the check into rules + review pack
Governance	3	Broker responsible; external DPO	Human approval stays; DPA with AI vendor

Conclusion: I&E is an early-stage AI adopter. The strategy must start small, use the data that already exists, need no IT team, and keep people in control.

3 AI vision and strategic goals

M1

Vision: "Every shipment is checked before it leaves. The AI does the searching; people keep the decisions and get their time back."

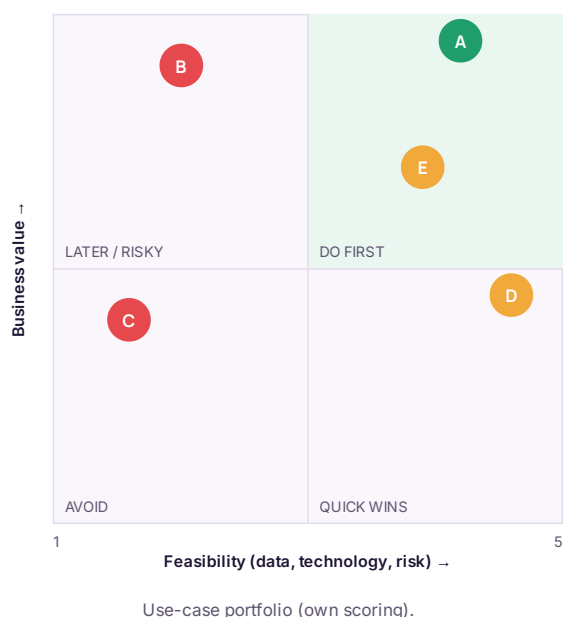
Strategic goal	KPI	Pilot target
Fewer border holds	Documentation-related customs holds	–50% vs. baseline
Time back for the team	Review minutes per shipment	< 10 min (from 30–45)
Reliable suggestions	HS codes accepted without change (6 digits, blind test)	≥ 90%
Safety first	False all-clears reaching customs	0 (hard limit)
Complete documents	Documents complete at first submission	95% (from ~70%)

Why AI now (M1): generative AI can now read free-text invoices in several languages and explain its answers; at the same time trade rules change faster than people can follow (CBAM, EU steel measure Reg. (EU) 2026/1384 with 50% out-of-quota duty and melt-and-pour proof from 1 Oct 2026, EUDR from 30 Dec 2026, e-invoicing 2027/28).

4 Finding and prioritising use cases

M6

Method: list the pain points from the client interview, turn them into candidate use cases, score each one 1–5 for **business value** and **feasibility** (data available, technology maturity, legal risk), then place them in a portfolio matrix.



Use case	Value	Feasib.	Decision
A Pre-shipment check + HS suggestion	4.8	4.2	Chosen (core)
E Tariff and rule-change monitor	3.8	3.9	Included as module
D Supplier e-mail triage + document request	2.8	4.6	Quick win, added (inbox, draft)
B Automatic customs filing	4.6	2.0	Ruled out: liability, trust
C Delay / ETA forecasting	2.6	1.6	Ruled out: no history data

Why A: high value (each hold costs about €1,500), feasible with existing documents, low risk because the AI only suggests and a person decides.

Why not B: the declarant is legally liable; automatic filing would also move the system towards a higher AI Act risk class.

Why not C: no historical delay data yet; possible after 12 months of logged decisions.

5 The selected solution: how the AI works

M2 · M4



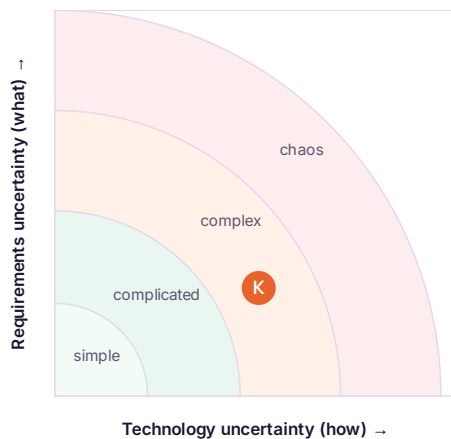
Around every step: Monitor (official EU/US sources, alerts) · review queue · LangSmith tests · decision log · Ask KlarSchiff assistant

KlarSchiff in five steps. Only step 3 uses generative AI; documents and legal consequences come from versioned rules.

Design choice	Why (link to the course)
RAG (retrieval over the 5,613 HS 2022 subheadings + reviewed knowledge base), not fine-tuning	No training data needed; answers stay linked to official sources; rules can be updated without retraining (M2, M4)
GPT-4o-mini , temperature 0, structured JSON output, model version pinned	Low cost (< €5/month in tokens), repeatable answers, evaluated before any model change
Rules engine separate from the LLM	Legal requirements must be checkable; the LLM only picks the code ("rules, not guesses")
Confidence threshold 0.75 + 13 review triggers	Unclear, translated, vague or Category 3 cases always go to a person
Human in the loop: approve / correct / reject	The broker stays responsible; every decision is logged
Course tools only: n8n, Python, OpenAI, LangChain/LangSmith, Streamlit, Airtable, Telegram	Built by a non-programmer consultant; easy to hand over
Answer cache , batch reuse, e-mail inbox, money tip (preferential origin), Ask KlarSchiff	Each one saves time or money; the assistant tells users it is an AI (AI Act Art. 50)

6 Implementation and operation

M7



Stacey matrix: KlarSchiff (K) sits in the **complex** zone.

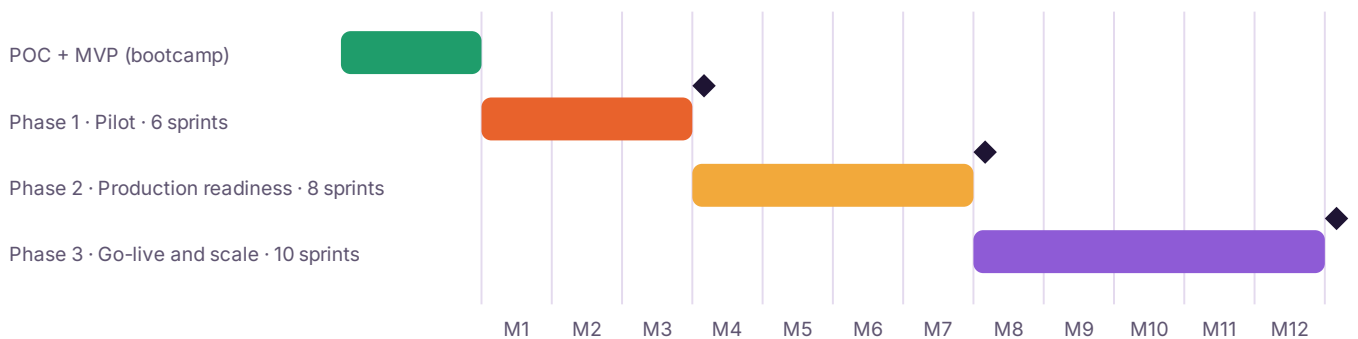
Why an agile method (Stacey matrix)

The **what** is fairly clear (check documents, suggest a code), but the **how** is uncertain: LLM behaviour, real document quality and changing trade rules. That places the project in the complex zone, where short feedback cycles work better than a fixed plan. So we use **Scrum** with 2-week sprints and a go / no-go gate after every phase.

Process model (CRISP-DM adapted to an LLM project)

Phase	KlarSchiff
Business understanding	Client interview, use-case matrix, KPIs
Data understanding	Sample invoices, HS 2022, official rule sources
Data preparation	Reviewed knowledge base, master list, rule files with dates
Modelling	Prompt + RAG + rules (no training)
Evaluation	20 + 17 test cases in LangSmith, LLM-as-judge, blind test
Deployment	Streamlit app, n8n workflows, monitor, review queue

Roadmap: about 12 months to full production



◆ = go / no-go gate. Benchmarks: brokerage AI pilots 10–12 weeks; enterprise AI rollouts 6–18 months; Gartner expects over 40% of agentic AI projects to be cancelled by the end of 2027, mostly for unclear value or weak risk controls.

Phase	Main work	Gate
1 · Pilot (months 1–3, 6 sprints)	Baseline · client data · blind test · shadow mode · live with review · decide	No false all-clear, ≥ 90% codes accepted, review < 10 min
2 · Production readiness (months 4–7, 8 sprints)	Security, EU hosting, ERP/TMS link, 10-digit TARIC, ATLAS-ready export for the broker, training	Security and data-protection sign-off
3 · Go-live and scale (months 8–12, 10 sprints)	Second route and product family, then customs brokers (one broker serves many importers)	KPIs stable for 3 months

Operation (run phase)

- **Monitoring:** daily rule monitor (EU and US sources) with alerts; every rule has a "last verified" date.
- **Quality:** weekly 10% second-look sample; corrections become new test cases; evaluation re-run before any model or prompt change.
- **Roles (Scrum):** product owner = I&E customs lead; Scrum master = consultant; team = consultant + client IT contact; broker = expert reviewer. Definition of Done: tested in LangSmith, reviewed by a person, documented, shown in the sprint review.

7 People, organisation and change

M5

Stakeholder	Interest	How we involve them
Chleo (CEO)	Value, risk, go / no-go	1-page monthly report, 30-min review, final pilot decision
Logistics team (2-3)	Faster checks, fewer surprises	2 key users design the review pack; 2-day training; weekly feedback
Customs broker	Clean review pack, legal responsibility	Labels the blind test; reviews Category 3 cases
Works council (if any)	No individual performance monitoring	Team-level metrics only (§87(1) no. 6 BetrVG), agreement before go-live
Data protection officer	GDPR	DPA with AI vendor, data minimisation, DPIA screening

- **AI literacy:** training on what the agent can and cannot do, automation bias and when to escalate (kept as a contract deliverable, even after the 2026 Digital Omnibus softened the wording of Art. 4).
- **Message to the team:** "It prepares, you decide." The goal is time back for work that needs people, not fewer people.
- **Against automation bias:** reasons and sources always visible; team-level override rate watched; weekly sample re-checked.

8 Governance, risk and compliance

M5 · M7

EU AI Act (Reg. (EU) 2024/1689, as amended by the Digital Omnibus Reg. (EU) 2026/1744)

- AI system: yes · prohibited practice: no · high-risk (Annex I / III): no, it is used by a private company and decides nothing about people → **minimal risk**.
- Art. 50(1) (since 2 Aug 2026): the Ask KlarSchiff assistant tells users they are talking to an AI.
- Re-check the class if the system ever files declarations, is used by an authority or evaluates employees.

GDPR

- Only goods descriptions go to the AI; scans: personal data covered before upload.
- E-mail inbox keeps sender data locally; retention 90 days; cache cleared at every model change.
- Processors: OpenAI, LangSmith (EU region), Airtable, Telegram, Google; SCC / EU-US Data Privacy Framework.

Top risk	Score (P×I)	Control
Tariff or rule change not noticed	16	Daily monitor, alerts, dates on every rule
Wrong HS code accepted	15	A person approves every code; threshold; Binding Tariff Information (vZTA) for recurring products
Automation bias	12	Visible reasons, override rate, spot audits
Personal data sent to an AI provider	12	Goods data only, DPA, EU region
Optimistic evaluation (same author wrote tests)	12	Blind test labelled by the broker
Liability for customs errors	10	Declarant stays responsible; contract and insurance

9 Business case and KPIs

M3 · M6

Cost item	Amount
Paid pilot (3 months)	€12,500 one-off
AI-literacy training	€1,300 one-off
Subscription from month 4	€3,500 / month
Technology (hosting, LangSmith, tokens)	~€150 / month

Value driver (base, per month)	Amount
Staff time saved (60 h × €40)	€2,400
Customs delays avoided (2 × €1,500)	€3,000
Fewer corrections and broker fees	€1,000
Total	€6,400

Scenario	Value / month	Break-even	ROI 12 m	ROI 36 m
Conservative	€3,200	none at €3.5k	-34%	-20%
Base	€6,400	month 7	+32%	+60%
Optimistic	€9,600	month 5	+99%	+141%

All values illustrative (client-interview exercise and own estimates); the pilot replaces them with measured data. The conservative case shows the price must follow measured value: a €2,000 tier breaks even in month 15.

Go / no-go after the pilot

- **Go if:** 0 false all-clears · $\geq 90\%$ codes accepted · review < 10 min · -50% holds trend.
- **Stop if:** a false all-clear reaches customs · more than 40% unnecessary reviews after tuning.

10 Results so far and lessons learned

M7 · REFLECTION

Built and tested in the bootcamp	Result
n8n proof of concept (Round 1)	Works end to end; demo video in the repository
Python agent + Streamlit MVP (EN/DE)	Check, review queue, batch, master list, tariff monitor, inbox, assistant, dashboard
LangSmith evaluation, 20 cases (v2.1)	HS code 0.95 · missing documents 1.00 · routed to a person 1.00 · no false all-clear 1.00
17 cases from other industries	Code right in 3 of 4; every unclear case sent to a person

Real error	Fix
E7 A vague shipment passed with no review	Vague text always goes to a person
E8 Right translation, wrong code	Close call → a person chooses
E9 A Turkish photo was labelled English	Own language check, re-tested
E10 Suggested a code that does not exist	Codes outside HS 2022 are flagged
E11 "Leistungserklärung fehlt" was not seen as missing	The check reads the whole word; re-tested, no regressions

- **No machine is perfect:** showing and fixing real errors builds more trust than a "zero errors" claim.
- **Honest numbers sell better:** the first payback estimate (1.6 months) ignored the subscription; the full model gives month 7.
- **Start small:** one route, one product family, a blind test first; scale only after a measured "go".

11 Sources

- IHK Akademie München, KI-Manager/-in (IHK): exam parts and 7 modules — ihk-akademie-muenchen.de/kuenstliche-intelligenz/ki-manager/
- IHK Trier, KI-Manager/in (IHK) course information 2026 (modules incl. Stacey matrix and process model)
- Regulation (EU) 2024/1689 (AI Act) and Digital Omnibus on AI, Regulation (EU) 2026/1744 (summary: Cooley, July 2026); Art. 50 status (Baker Botts, Sept 2026)
- Regulation (EU) 2026/1384 (EU steel measure), application 1 July 2026 (Reed Smith, June 2026)
- Regulation (EU) 2023/1115 (EUDR) as amended: 30 Dec 2026 / 30 Jun 2027 (Stibbe)
- Regulation (EU) 2023/956 (CBAM) and 2025 simplification; Construction Products Regulation (EU) 2024/3110
- Gartner press release, June 2025: over 40% of agentic AI projects will be cancelled by the end of 2027
- Scrum Guide 2020 (Schwaber & Sutherland); CRISP-DM process model
- Project repository: [use_case_definition.md](#), [roi_risk_assessment.md](#), [strategic_plan.md](#), [compliance/eu_ai_act_compliance.md](#), [compliance/gdpr_documentation.md](#), [mvp/mvp_documentation.md](#)